

Design of Mentally and Physically Demanding Tasks as Distractors of Rotation Gains

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ABSTRACT

Redirected walking enables natural walking in virtual reality under the constraints of limited physical space by manipulating position and orientation of the user. One such manipulation method are rotation gains which decouple real and virtual head turning. When reaching a boundary of the tracking space, the user's orientation can be reset by applying a rotation gain and forcing the user to do a certain task that requires head rotations. To identify what kind of tasks are best suited to mask the redirection, four tasks were designed that differentiate in their amount of mentally and physically demand: a spatial memory task, a verbal memory task, a physically exhausting task and a task which requires physical skill. A first pilot study was conducted to evaluate the influence that the different task categories have on the redirection awareness. The results showed overall low redirection detection rates, with the physically exhausting task being the least noticeable, whilst the physical skill task was detected the most, and the mentally demanding tasks in between. Although no clear distinction regarding the redirection awareness between mentally and physically demanding tasks could be shown, the insights gained from this work and the study, can be used to improve future implementations of rotation tasks.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality;

1 INTRODUCTION

Rotation gains are subtle manipulations of the user's rotation [4] that can be used whenever the user is about to trespass their real-world boundaries. This can be achieved by having an event or task that forces the user to rotate in place while their virtual rotation is amplified until the desired redirection is reached. For the rest of this work, we are going to name these tasks *rotation tasks*, as they have to incorporate an action or incentive for the user to rotate.

Research efforts have gone into determining better strategies to mask the redirection. For instance, through the use of distractors [2] and interactions [1]. While these already give lots of insights into the possible design of rotation tasks, there are still more layers to explore in order to create a better experience for applications that have to make use of rotation tasks. One such unexplored layer is the difference in mental and physical demand that tasks put upon the user.

The goal of this work is therefore to design rotation tasks that vary in their physical and mental demand on the user and evaluate their influence on altering the user's awareness of the applied rotation gain.

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Figure 1: Whac-A-Mole task: arcade cabinet with several holes.

2 ROTATION TASKS

The rotation tasks were designed with certain premisses in mind. Similar to the approach of Sra et al. [3], the rotation tasks should be integrated into every aspect of the experience, from the environment to the user's activity and also the narration. The tasks also need to be designed in a way that makes sure that the user cannot finish the task if not fully redirected. Otherwise, the user would be able to proceed with a play space rotation that does not fit the upcoming level design any more. Most of the presented tasks, will therefore be potentially endless or based on a system that couples the redirection to the task progression, which in turn ensures a complete redirection before the task is finished.

The developed tasks include:

• Physically Demanding Tasks

- **Physical Skill: Whac-A-Mole** The user is presented with a Whac-A-Mole arcade cabinet (a reaction time game) and tasked to use the hammer and hit the moles, that periodically pop out of the holes, before they disappear again. Every successful hit increases the user's score, which is displayed right in front of them (see Figure 1).
- **Physical Exhaustion: Pressure Management** The user is assigned to take care of the pressure cycle in the control room of a factory (see Figure 3). First he has to operate a pump, positioned on the right side, by alternating a downwards push between his two hands. Afterwards, the excess pressure from pumping can be regulated by turning the valve on the left side. As soon as the pressure reaches the optimal value, the regulator handles on the front wall, get sucked in and the user has to pull them out again.

• Mentally Demanding Tasks

- **Spatial Memory: Painting Restoration** The user has to scan a specific area for a predefined cue. In order

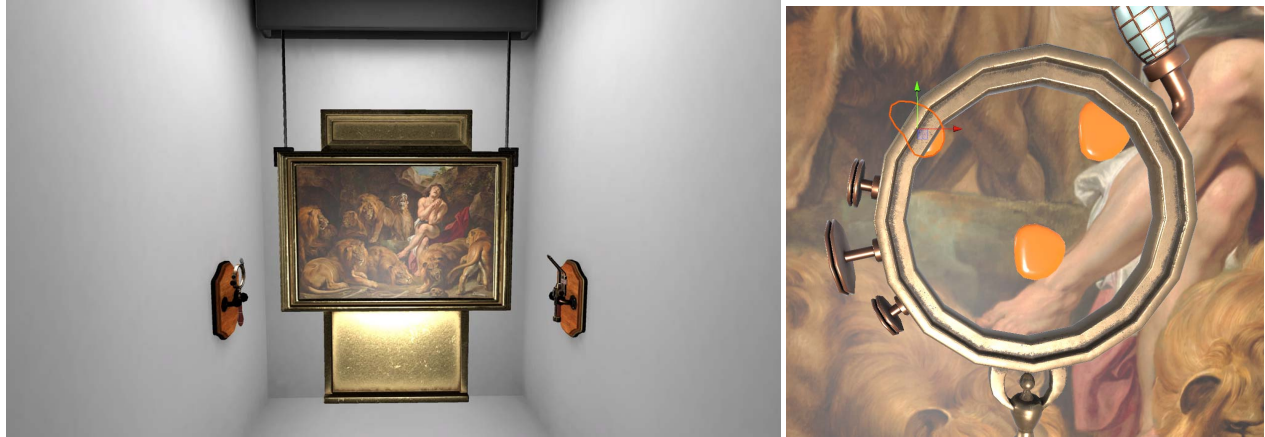


Figure 2: Left: Painting restoration task: Lens tool to the left, remover tool to the right, and a painting in the middle. Right: Blobs of the substance that has to be removed. They are only visible when using the lens tool.

to fulfil the task, the user has to remember the location of the found cue and remove it with a given tool (see Figure 2).

- **Verbal Memory: Chemical Lab** In front of the user, a display hangs on the wall, showing an arithmetic problem. Depending on the type of calculation, the properties need to be added, subtracted, multiplied or divided. The user is assigned to solve the problem and choose the chemical bottle with the respective number written on it. The bottles are stored on shelves to the left and right of the user and numbered from five to twenty-eight (see Figure 4).

3 USER STUDY

The designed rotation tasks were evaluated in a preliminary pilot study with 8 participants to test the influence of mentally and physically demanding tasks on detection of rotation gains. Moreover, it was examined if the tasks are indeed physically, or alternatively, mentally demanding.

Results suggest that the physically demanding tasks are indeed physically demanding. For the mentally demanding tasks, only the verbal memory based task was rated more mentally demanding, while the spatial memory task could not be differentiated from the physical tasks in terms of the mental demand. In regard to the masking of the redirection, all tasks managed a subtle rotation of the user, with all of them having a detection rate below the probability of simply guessing, which would be at 50%. However, in comparison, the task requiring physical skill performed the worst, while the physical exhaustion task had the lowest detection rate. For the mental tasks, the difference between them was smaller and both showed good performances.

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Figure 3: Pressure management task: a valve to the left, levers in the front, and a pump to the right.



Figure 4: Chemical lab task: Bottles to the left and right, a screen and a transporter table in the center.